I. Data type

1. Numeric Data Types

Python numbers represent data that has a numeric value. A numeric value can be an integer, a floating number or even a complex number. These values are defined as int, float and complex classes.

- **Integers: value is represented by int class. It contains positive or negative whole numbers (without fractions or decimals). There is no limit to how long an integer value can be.**
- **Float: value is represented by float class. It is a real number with a floating-point representation. It is specified by a decimal point. Optionally, character e or E followed by a positive or negative integer may be appended to specify scientific notation.**
- **Complex Numbers: It is represented by a complex class. It is specified as (real part) + (imaginary part)j. For example - 2+3j**

2. Sequence Data Types

A sequence is an ordered collection of items, which can be of similar or different data types. Sequences allow storing of multiple values in an organized and efficient fashion. There are several sequence data types of Python:

String Data Type

Python Strings are arrays of bytes representing Unicode characters. In Python, there is no character data type, a character is a string of length one. It is represented by str class.

Strings in Python can be created using single quotes, double quotes or even triple quotes. We can access individual characters of a String using index.

List Data Type

Lists are similar to arrays found in other languages. They are an ordered and mutable collection of items. It is very flexible as items in a list do not need to be of the same type.

Tuple Data Type

Tuple is an ordered collection of Python objects. The only difference between a tuple and a list is that tuples are immutable. Tuples cannot be modified after it is created.

3. Boolean Data Type

Python Boolean Data type is one of the two built-in values, True or False. Boolean objects that are equal to True are truthy (true) and those equal to False are falsy (false). However non-Boolean objects can be evaluated in a Boolean context as well and determined to be true or false. It is denoted by class bool.

4. Set Data Type

In Python Data Types, Set is an unordered collection of data types that is iterable, mutable, and has no duplicate elements. The order of elements in a set is undefined though it may consist of various elements.

5. Dictionary Data Type

A dictionary in Python is a collection of data values, used to store data values like a map, unlike other Python Data Types, a Dictionary holds a key: value pair. Key-value is provided in dictionary to make it more optimized. Each key-value pair in a Dictionary is separated by a colon : , whereas each key is separated by a 'comma'.

II. I/O in python

Taking input in Python

Python's input() function is used to take user input. By default, it returns the user input in form of a string.

Printing Output using print() in Python

At its core, printing output in Python is straightforward, thanks to the print() function. This function allows us to display text, variables and expressions on the console.

Printing Variables

We can use the print() function to print single and multiple variables. We can print multiple variables by separating them with commas.

Take Multiple Input in Python

We are taking multiple input from the user in a single line, splitting the values entered by the user into separate variables for each value using the split() method. Then, it prints the values with corresponding labels, either two or three, based on the number of inputs provided by the user.

Change the Type of Input in Python

By default input() function helps in taking user input as string. If any user wants to take input as int or float, we just need to typecast it.

III. Operator

In Python programming, Operators in general are used to perform operations on values and variables.

- **Operators:** Special symbols like -, + , * , /, etc.
- **Operands:** Value on which the operator is applied.

Types of Operators in Python

Operators in Python

Type	Operators
Arithmetic Operators	+, -, *, /, %
Relational Operators	< , <= , > , >=, ==, !=
Logical Operators	AND, OR, NOT
Bitwise operator	&, |, ^, ~, <<, >>
Assignment Operator	= , += , -= , *= , %=

Arithmetic Operators

Python Arithmetic operators are used to perform basic mathematical operations like addition, subtraction, multiplication and division.

In Python 3.x the result of division is a floating-point while in Python 2.x division of 2 integers was an integer. To obtain an integer result in Python 3.x floored (// integer) is used.

Comparison Operators

In Python, Comparison (or Relational) operators compares values. It either returns True or False according to the condition.

Logical Operators

Python Logical operators perform **Logical AND**, **Logical OR** and **Logical NOT** operations. It is used to combine conditional statements.

The precedence of Logical Operators in Python is as follows:

1. Logical not
2. logical and
3. logical or

Bitwise Operators

Python Bitwise operators act on bits and perform bit-by-bit operations. These are used to operate on binary numbers.

Bitwise Operators in Python are as follows:

1. **Bitwise NOT**
2. **Bitwise Shift**

3. **Bitwise AND**
4. **Bitwise XOR**
5. **Bitwise OR**

Assignment Operators

Python Assignment operators are used to assign values to the variables. This operator is used to assign the value of the right side of the expression to the left side operand.

Identity Operators

In Python, **is** and **is not** are the identity operators both are used to check if two values are located on the same part of the memory. Two variables that are equal do not imply that they are identical.

is *True if the operands are identical*
is not *True if the operands are not identical*

Membership Operators

In Python, **in** and **not in** are the membership operators that are used to test whether a value or variable is in a sequence.

in *True if value is found in the sequence*
not in *True if value is not found in the sequence*

Ternary Operator

in Python, Ternary operators also known as conditional expressions are operators that evaluate something based on a condition being true or false. It was added to Python in version 2.5.

It simply allows testing a condition in a **single line** replacing the multiline if-else, making the code compact.

Precedence and Associativity of Operators

In Python, Operator precedence and associativity determine the priorities of the operator.

perator Associativity

If an expression contains two or more operators with the same precedence then Operator Associativity is used to determine. It can either be Left to Right or from Right to Left.